Yuriel Wang Jun Long Ryan

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Education

Singapore University of Technology and Design (SUTD) 2025 – 2026
MEng (Research), Artificial Intelligence

- cGPA: *5.00/5.00*
- Funded by AI Singapore Accelerated Masters Scholarship (2024 - 2026)
- Funded by DSO-AISG Incentive (Research) Award (2025-2026)

Singapore University of Technology and Design (SUTD) 2021 – 2025
BEng (Computer Science) | Minor in Artificial Intelligence

- cGPA: 4.70/5.00, Honours with Highest Distinction
- Funded by SUTD Undergraduate Merit Scholarship (2021 - 2024)

Scholarships & Awards

- *DSO-AI Singapore Incentive (Research) Award* (2025)
- *1st Runner-Up, National AI Student Challenge* (TikTok Track) (2025)
- *SUTD Honours List* (2025)
- *AI Singapore Accelerated Masters Program* (2024)
- *Jyoti and Aditya Mathur Student Achievement Award* (2024)

Publications

Self-Captioning Multimodal Interaction Tuning: Amplifying Exploitable Redundancies for Robust Vision Language Models *Jul 2026*
*International Conference on Machine Learning (**ICML** 2026, Main Conference, Poster)*
Yuriel Ryan, Ip Hei Man, Adriel Kuek, Paul Pu Liang, Roy Ka-Wei Lee

Balance Human Agency & AI Assistance in the Tussle for the "Right" to Choose, Own, Work, and Learn *Jul 2026*
*Trustworthy AI for Good Workshop (**ICML** 2026, Workshop, Poster)*
Zi-Yu Khoo, **Yuriel Ryan***, Nicole Heng Yim Oo*, Hui En Pang, Eric J. W. Orlowski, Hakim Norhashim, Davin Choo, Ruth Wan Theng Chew, Rachael Hwee Ling Sim, Simon Chesterman, Jungpil Hahn, Bryan Kian Hsiang Low

Large Scale Narrative Analysis of Multimodal Memes *May 2026*
*Proceedings of the International AAAI Conference on Web and Social Media (**ICWSM** 2026, Main Conference, Poster)*
Jia Wang Peh , Ming Shan Hee , Bryan (Chen Zhengyu) Tan, **Yuriel Ryan**, Roy Ka-Wei Lee

Humor in Pixels: Benchmarking Large Multimodal Models Understanding of Online Comics *Nov 2025*
*Association for Computational Linguistics (**EMNLP** 2025, Findings, Poster)*
Yuriel Ryan*, Rui Yang Tan*, Kenny Tsu Wei Choo, Roy Ka-Wei Lee

* *Denotes equal contribution.*

Work Experience

Research Intern, DSO - AISG Research Award | [GitHub] Sep 2025 – Present

- Collaborated with research scientists from DSO National Laboratories as part of the research award to improve VLM robustness against ambiguous and corrupted modalities.
- Applied the Partial Information Decomposition (information theoretic) framework to motivate hypotheses and design experiments for tuning multimodal interactions.
- Curate training (e.g. cleaning, deduplication) and validation partitions using **vLLM**, yielding three training sets (984,000 samples) of varying redundant interactions for supervised fine-tuning.
- Tested the hypotheses by fine-tuning VLMs—ranging from 256M (SmolVLM) to 8B (LLaVa-OneVision) parameters—with low rank adapters, leading to a 38.3% decrease in visual-induced hallucinations and 16.8% gain

in consistency; methods and findings were accepted into the ICML main track.

Founding AI Engineer, Pallo (Iterative W25) Jan 2025 – Mar 2025

- Built a RAG system to ground large language models (LLMs) on Singapore exam syllabi, securing **pre-seed funding** from Iterative VC (Winter 2025).
- Built a document processing pipeline with open-source Vision Language Models, contributing 8,700 high-quality questions to enable RAG.
- Deployed **DeepSeek R1 models** to **Google Cloud Run** combined with RAG for solving Singapore GCE A-Level math problems, increasing accuracy scores by 36%.

Research Assistant, Social AI Lab (SUTD) June 2024 – Dec 2024

- Automated scripts with **Selenium** and **Beautiful Soup** to collect 28,000 web comics for analysis.
- Recruited and managed 8 participants to annotate 2,800 comics with Label Studio to curate a benchmark.
- Evaluated large vision language models (e.g., Qwen2-VL-72B) to identify weaknesses and biases in sequence recognition and humor comprehension, leading to a publication in EMNLP Findings.
- Maintained a GitHub codebase for the collected data and ensured reproducible pipelines for evaluating VLMs on PixelHumor.

Relevant Projects

Vision & Text Modalities | Augment and Think Before Answering: VLM Agentic Workflow [GitHub], 2025

- Designed an agentic workflow to integrate external metadata (e.g., user comments) and slow down video inputs, exposing fine-grained temporal cues often missed by fixed-rate sampling.
- Implemented structured chain of thoughts using Google Gemini Pro 2.5 that enforces an "Augment Think (ANT) Before Answering" strategy to mitigate misleading queries.
- **Outperformed GPT-4o** across multiple question types, achieving a 58.07% Correctness Score (vs GPT-4o 45.2%) and a 15.3% Robustness Score (vs GPT-4o 8%) through the proposed Augment and Think pipeline, securing the **1st Runner-Up** award in the National AI Student Challenge (TikTok track).

Audio Modality | D'Noise (Capstone Project): Speech Enhancement [GitHub], 2025

- Trained Google's Guided Speech Enhancement (GSE) Network in **PyTorch** to achieve an 87.2% noise reduction over non-deep learning methods with real-time inference (20ms latency).
- Collected over 20 hours of beamformed audio with IMDA's Singaporean local English to train the GSE Network with **Google Cloud Platform's Compute Engine** and **Cloud Storage**.

Visual Modality | No Pain, Just Gain (50.035 Computer Vision): **Pose Estimation** [GitHub], 2024

- Trained Google's BlazePose model in **TensorFlow**, introducing Convolutional Block Attention Modules (CBAM) to achieve near real-time inference (100ms) for estimating **skeleton keypoints** in human poses.

Thermal & Depth Modalities | SmartDrive (UROP): Driver Fatigue Detection 2023

- Implemented a low-light fatigue detection pipeline using Luxonis Depth Cameras and FLIR Thermal Sensors through vision-based algorithms: PERCLOS and Head Pose Estimation.

Additional Information

Research Interests: multimodal machine learning, multimodal interactions, and information theory.
Technical Skills: Python, PyTorch, TensorFlow; Hugging Face Transformers, vLLM, LangChain; LoRA / parameter-efficient fine-tuning, retrieval-augmented generation, multimodal instruction tuning; Google Cloud Platform (Compute Engine, Cloud Run, Cloud Storage), Git.
Certifications: Google Professional Machine Learning Engineer
Community Involvement: Singapore Youth AI Sub-committee | Access Singapore Volunteer | SUTD Climbers Secretary | Freshman Orientation Welfare Executive | TA for SUTD Freshmore Math and Computing courses.
Languages: Native English (spoken and written) | Conversational Mandarin (spoken)